Carbon Credit Auction Clearing via Ising Optimisation

Quantum Dice Trinity Challenge 2026 Stage 1 Proposal by Ballu Di Team

1. Problem Context

Carbon credit allocation is the primary mechanism by which regulators distribute emissions allowances to industry. Under the EU ETS, the European Commission distributes a fixed system-wide cap across heterogeneous credit categories: general allowances (EUA) for stationary and maritime installations, aviation allowances (EUAA), member-state sub-quotas for 25 participating states, and dedicated pools for the Innovation, Modernisation, and Recovery and Resilience Funds. Each category carries distinct eligibility rules and sector-specific supply constraints not resolved by the uniform-price auction mechanism alone.

The allocation problem is combinatorial: N demands compete for M credit lots differentiated by *credit type*, *sector eligibility*, and *fund designation*. Accepting one demand blocks others requesting overlapping lots under the system-wide cap. This is weighted k-set packing, where k is the maximum lot types in any single demand. Hazan, Safra and Schwartz [1] proved no polynomial-time algorithm approximates this within $o(k/\log k)$ unless P = NP, motivating a probabilistic computing approach.

2. End-User and Operational Grounding

The solution targets European Commission allocation designers resolving cross-category distribution of the 2024 system-wide cap of 1.386 billion allowances across EUAs, EUAAs, and member-state sub-quotas [5]. From 2025, EUAA volumes are integrated into EUA spot auctions, creating explicit cross-type allocation decisions requiring combinatorial treatment. All inputs demand values v_i, lot matrices a_{ij}, per-type supply S_τ, and type assignments $\tau(i)$ are determined by published regulatory decisions with no estimation required; integer programming serves as the natural classical baseline.

3. Ising Formulation

Taking the standard auction clearing setup of binary variables $x_i \in \{0, 1\}$, welfare objective $\max \sum_i v_i x_i$, and lot exclusivity constraints $\sum_i a_{ij} x_i \le 1$ as given, we encode the full problem as an Ising ground-state search following Lucas [2]. Each x_i maps to a spin $\sigma_i \in \{-1, +1\}$ via $x_i = (1 + \sigma_i)/2$, where $\sigma_i = +1$ denotes acceptance.

Ising Hamiltonian

The total Hamiltonian $H = H_{\text{obj}} + H_{\text{excl}} + H_{\text{cap}}$ is:

$$H_{\text{obj}} = -A \sum_i v_i x_i \tag{1}$$

$$H_{\text{excl}} = B \sum_j \sum_{i<k} a_{ij} a_{kj}\, x_i x_k \tag{2}$$

$$H_{\text{cap}} = C \sum_\tau \left(\sum_{\{i:\tau(i)=\tau\}} x_i - S_\tau \right)^2 \tag{3}$$

H_{obj} encodes the welfare reward: spins with large v_i experience a strong external field biasing them toward $\sigma_i = +1$. H_{excl} encodes lot exclusivity: any two demands sharing lot j incur an antiferromagnetic coupling, penalising simultaneous acceptance. H_{cap} enforces per-type supply, penalising over-allocation of each credit type τ against its supply S_τ independently, so that a penalty on EUA lots does not interact with the EUAA cap. Substituting $x_i = (1 + \sigma_i)/2$ and collecting terms yields the standard Ising form:

$$H = -\sum_i h_i \sigma_i - \sum_{i<k} J_{ik}\, \sigma_i \sigma_k + \text{const} \tag{4}$$

where h_i encodes per-demand reward and J_{ik} encodes pairwise conflict strength. The ground state of H corresponds to the optimal allocation, found by the ORBIT simulator via stochastic spin-flip dynamics.

Penalty Conditions

For the ground state to be strictly feasible, each penalty must dominate the objective under any single violation:

$$B > 2A \cdot \max_i (v_i) \tag{5}$$

$$C > A \cdot \max_i (v_i) \tag{6}$$

Condition (5) ensures no conflicting demand pair is jointly accepted regardless of combined value. Condition (6) ensures exceeding any per-type supply by one unit always costs more than the value of that demand. Setting $A = 1$, $B = 2 \max_i(v_i) + 1$, and $C = \max_i(v_i) + 1$ satisfies both at the tightest feasible bound; practical guidance on coefficient calibration is in [3].

4. Modelling Trade-offs

Hard vs. soft constraints. Lot exclusivity is a hard regulatory rule requiring strict dominance (5). Per-type caps (6) may be treated as soft where the Market Stability Reserve permits inter-category rebalancing, by reducing C accordingly.

Per-type cap approximation. H_{cap} treats each type τ as having independently fixed supply S_τ. In practice EU ETS supply is subject to Market Stability Reserve adjustments not always pre-assigned by type before allocation. The penalty remains exact where supply is fixed in published regulatory decisions.

Realism vs. tractability. Package complementarities introduce degree-p terms for p-lot bundles, reducible to QUBO via Boros and Hammer [4] at the cost of $p - 2$ auxiliary variables per bundle, expanding variable count from N to $O(N\bar{p})$. We adopt the independent valuation assumption in the base formulation, identifying this as the primary extension.

References

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[6] P. Cramton, Y. Shoham, and R. Steinberg, *Combinatorial Auctions*, MIT Press, 2006.